

Dense Language Rewards for Reasoning

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1 Experiments

Setup: We fine-tune Qwen2.5-7B-Instruct on GSM8K using GRPO, comparing two reward conditions over 200 training steps. The binary baseline assigns $r = 1$ if the final answer is correct and $r = 0$ otherwise. Our proposed method, DLRR (Dense Language Rewards for Reasoning), replaces the binary signal with step-level scores from a GPT-4o-mini critic. For each completion, the last five reasoning steps are extracted and scored on two dimensions: correctness $c \in \{0, 0.5, 1\}$ and calibration $k \in \{-0.2, 0, 0.1\}$, where calibration penalizes overconfident language on incorrect steps and rewards well-justified certainty. Step scores are aggregated as:

$$\frac{\sum_{t=1}^T w_t(c_t + k_t)}{\sum_{t=1}^T w_t}, \quad w_t = \begin{cases} 2 & t = T \\ 1 & \text{otherwise} \end{cases}$$

Both runs use identical hyperparameters: LoRA ($r = 64$, $\alpha = 128$, all attention and MLP projection layers), 4-bit NF4 quantization, learning rate $\eta = 10^{-6}$, batch size 1, gradient accumulation steps 8, and 2 generations per prompt. GSM8K test accuracy is evaluated on 100 held-out examples every 50 training steps.

Table 1: GSM8K held-out accuracy (100 examples) at each evaluation checkpoint.

Method	Base	Step 50	Step 100	Step 150
Base model (no tuning)	61%	61%	61%	61%
Binary GRPO	61%	71%	73%	74%
DLRR (ours)	61%	60%	64%	65%

Initial Results: The binary baseline achieves 71%, 73%, and 74% accuracy at steps 50, 100, and 150 respectively, compared to a base model accuracy of $\approx 61\%$ demonstrating consistent improvement under sparse outcome supervision. DLRR achieves 60%, 64%, and 65% at the same checkpoints which is lower than the binary baseline despite a higher training reward mean (0.85-0.88 for DLRR vs 0.65-0.80 for binary baseline). This discrepancy between training reward and held-out accuracy suggests the DLRR critic may be rewarding well-structured reasoning independently of correctness, a form of reward misalignment in which the model learns to produce calibrated-sounding steps without improving answer accuracy. Notably, both methods show consistent upward accuracy trends over training, and DLRR’s absolute improvement of 5% over 150 steps is comparable to the binary baseline’s 3%, suggesting the two methods may converge with additional training.

2 Changes to Research Hypothesis or Objective

Original Hypothesis: Step-level natural language feedback from an LLM critic will provide a denser reward signal than binary outcome supervision, improving sample efficiency and final accuracy on GSM8K under a fixed training budget.

Revised Hypothesis: Our hypothesis is refined in light of initial results. While DLRR demonstrates a consistent upward accuracy trend comparable to the binary baseline, the higher training reward does not translate directly to higher held-out accuracy at 150 steps, suggesting that the calibration component of the reward signal may be introducing misalignment. We now hypothesize that step-level correctness feedback alone will outperform binary reward, but that the calibration signal as currently formulated may inadvertently reward fluent but incorrect reasoning. The revised objective is to isolate these two effects via ablation and, if necessary, recalibrate the reward to more tightly couple critic scores with ground-truth correctness.

Justification: The gap between DLRR training reward (0.85–0.88) and held-out accuracy (60–65%) is the key empirical signal driving this revision. A well-aligned reward signal should correlate positively

with held-out accuracy; the observed divergence is consistent with reward hacking, where the model optimizes for critic approval rather than correctness. This motivates both the correctness-only ablation and a potential hybrid objective

$$R_{\text{combined}} = \lambda \cdot R_{\text{DLRR}} + (1 - \lambda) \cdot \mathbf{1}[\text{correct}]$$

as a mitigation strategy.

3 Next Steps

1. **Further Experimentation:** Run a correctness-only ablation of DLRR ($k_t = 0$ for all steps) to isolate whether reward misalignment stems from the calibration signal specifically or from step-level scoring in general. Collect step 200 accuracy for both current runs to complete Table 1.
2. **Algorithm Refinement:** If the ablation confirms calibration-induced reward hacking, adopt the hybrid objective mentioned above. to anchor step-level scores to ground-truth correctness.
3. **Evaluation & Analysis:** Evaluate all three conditions (binary GRPO, DLRR, correctness-only ablation) on the full GSM8K test set. Manually inspect high-scoring DLRR completions at steps 50–150 to confirm whether reward hacking is occurring, and report final accuracy alongside reward curves in the paper.

Potential Challenges: The primary risk is reward hacking, suggested by the divergence between DLRR training reward (0.85–0.88) and held-out accuracy (60–65%). A secondary risk is GPT-4o-mini critic inconsistency, which could be mitigated by ensemble scoring or replacing arithmetic verification steps with a deterministic rule-based checker.

Team Contributions:

- **Diya Kejriwal:** Led infrastructure and training pipeline development (Modal, W&B, HuggingFace), implemented and debugged GRPO, and ran baseline/DLRR experiments; will complete the correctness-only ablation, fix evaluation logging, and run final GSM8K evaluations.
- **Gurmeher Kaur:** Designed and implemented the DLRR reward function (step splitting, GPT-4o-mini critic, calibration scoring) and conducted literature review; will implement and tune the hybrid objective and address critic inconsistency via ensemble scoring.
- **Andrew Su:** Led experimental design, hypothesis framing, and initial analysis (including reward hacking diagnosis) and wrote the milestone report; will conduct qualitative analysis of DLRR outputs and complete the final paper and results.

References

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